MIL-E-1/33C
1 [illegible] 1968
SUPERSEDING
MIL-E-1/33B
21 January 1959

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 3A5

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Twin triode, miniature, high frequency

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated filament

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	-f (see note 1)	2p	2g	fct (see note 1)	1g	1p	+f (see note 1)

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ik/k	Pp	Alt
Unit:	Vdc	Vdc	Vdc	mAdc	W	ft
Maximum:	1.4 or 2.8 (±15%)	150	---	14	1.0	(see note 5)
TEST CONDITIONS:	1.4	135	-1.5	---	---	---

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required

E-50.2 Holding period

E-50.3 Preheating

5. Preparation for delivery

© denotes changes

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1301	Filament current		0.65	II	If	200	240	mA
1266	Total grid current	See notes 1 and 6	0.65	II	Ic	0	-1.5	μAdc
1256	Electrode current (1) (anode)	See note 3	0.65	II	Ib	8.3	16.7	mAdc
1236	Power oscillation (2)	Power oscillation (1); Ef = 1.1 Vdc (see note 6)	0.65	II	Po	0.45	---	W
1201	Short and discontinuity detection	See note 7	0.4	II	---	---	---	---
Appendix D, 20(a), 30	Visual and mechanical inspection criteria		---	---	---	---	---	---
Appendix D, 20(b)	Dimensions	Outline 5-2, dimensions A, C, D, and E	1.0	I	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration	Rp = 2,000 ohms; (see note 2)	6.5	L6	Ep	---	600	mVac
1256	Electrode current (2) (anode)	Ec = -10.5 Vdc; Eb = 90 Vdc (see note 4)	6.5	L6	Ib	---	375	μAdc
1306	Transconductance	See note 3	6.5	L6	Sm	2,080	3,120	μmhos
1316	Amplification factor	See note 3	6.5	L6	Mu	13	17	---
1211	Insulation of electrodes		4.0	L6	R	10	---	Meg
1236	Power oscillation (1)	F = 50 MHz; Push-pull; Ib = 30 mAdc; Ic = 6 mAdc; Rg = 4,000 ohms	6.5	L6	Po	1.4	---	W
1331	Direct-interelectrode capacitance	Without shield	6.5	L6	Cgp Cin Cout	2.7 0.70 0.70	3.7 1.10 1.30	pf pf pf
1121 and 4.3.2	Base strain		---	---	---	---	---	---
2126	Glass strain		4.0	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---
	Quality conformance inspection, part 3							
4.7 and 1501	Life-test provisions (intermittent)	Group A: Ef = 1.4 Vdc or Vac with equivalent Ec	---	---	---	---	---	---
4.7.3	Life-test end point (intermittent) (250 hours)	Transconductance	---	---	Sm	1,690	---	μmhos

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NOTES:

1. For parallel filament operation, pin number 4 is positive, pin numbers 1 and 7 are negative.
2. Tie anode 1 to anode 2; tie grid 1 to grid 2 at socket.
3. Test each unit separately.
4. Test each unit separately. Unit not under test, Ec = -50 Vdc.
5. See 6.5.2.3 for reduced pressure (altitude), and for altitude, maximum peak voltage. ©
6. This test to be performed at the conclusion of the holding period.
7. Test equipment shall be ac operated, and shall be capable of indicating tap shorts of 50,000 ohms, or less.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL
Navy - SH
Air Force - 11, 85
DSA - ES

User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 19

Preparing activity:
Navy - EC

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